XRPL Validator — Daily Engineering Report · 2026-10-16

Generated 2026-10-17 00:20 EDT. Scope: XRPL work only — the git history of xrpl-validator and the XRPL Rust SDK, the live native-engine shadow, the reference rippled node, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **2 commits** (xrpl-validator 2)
- **1 engine finding fixed** (F421)
- Live shadow: 8,019/8,019 ledgers fully matched, 999,050 transactions, 0 engine receipts in the current soak
- Amendment **fixCleanup3_4_0** held majority (since 2026-10-09 09:34 UTC); activation due ~2026-10-23 09:34 UTC

Live validator and native engine

Snapshot at generation time.

Measure	Value
Ledgers compared (current run)	8,019
Ledgers fully matched	8,019
Transactions applied natively	999,050
Result mismatches	0
Byte mismatches	0
Missing / extra keys	0 / 0
Batch inner-result mismatches	0
Engine receipts (current soak)	0
Live engine build running since	2026-10-16 15:38 EDT
Reference rippled	3.4.1, full, validated #107,709,662

Engine findings fixed

- **fix(ledger): F421 — a crossed offer's remainder is repriced through the native rate (a 17-digit TakerPays is not shrunk)** (`fbcc3ae`, xrpl-validator) flowCross reprices the rested remainder through Quality{takerAmount.out, takerAmount.in}.rate(), and rippled's divide never shrinks a native mantissa (Finding 240): with a seventeen-digit drop count the quotient is sixteen digits and the +5 lands on its last digit. The book page already used keylet::rate_encode_native; the three reprice sites in the placement block still used rate_of_me (the IOU rule), so the …

Engineering changes

- **fix(node): Step 3 salts with the set more than half the trusted list names, not only a full quorum** (`b6948f4`, xrpl-validator) Mainnet #107700728 (2026-10-16, soak #55): only 27 trusted validations of quorum 28 reached us on #107700727-734, so agreed_set gave None and Step 3 salted with the ledger's own root (salt_own=8). On #107700728 the consensus set held a transaction the ledger never filed, so our order differed at position 0 (382 = 382 transactions, 4 results and 57 objects differed). The 27 that did reach us all named the network's …

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_4_0	2026-10-09 09:34 UTC	2026-10-23 09:34 UTC

Amendments enabled on mainnet: 97.

Appendix — commits

Time	Repo	Commit	Subject
09:43	xrpl-validator	`fbcc3ae`	fix(ledger): F421 — a crossed offer's remainder is repriced through the native rate (a 17-digit TakerPays is not shrunk)
14:54	xrpl-validator	`b6948f4`	fix(node): Step 3 salts with the set more than half the trusted list names, not only a full quorum